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Defining and quantifying oxygen delivery potency of blood products

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A complete list of the members of the CONCERT Study Group appears in the supplemental Appendix.

The full-text version of this article contains a data supplement.

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Abstract

Transfusion is intended to correct anemia when it is believed to either impair adequate oxygen (O₂) delivery or when the resulting compensatory cardiovascular workload poses clinical risk. Dosing is typically based on hemoglobin (Hb) levels, without consideration of differences in red blood cell (RBC) or Hb-based blood substitute O₂ delivery potential. Such consideration requires quantified assessment of O₂ delivery potency, for which there is no accepted metric. We developed an in vitro metric to predict the O₂ delivery potency for various “Hb formulations” (fresh and stored whole blood [sWB]), stored RBC concentrate (sRBCc), and artificial WB analogs (WBAs) composed with different Hb-based O₂ carriers. O₂ uptake and delivery were modeled across physiologic O₂ gradients by integrating data from matched O₂ association (high pH; ie, in the lungs) and dissociation curves (low pH; ie, in tissues). These data were used to compute a novel O₂ delivery potency metric, lung-to-tissue O₂ flux (L-TOF). Using L-TOF, we quantified O₂ delivery potency of sRBCc (additive solution 1) and determined that equipotent transfusion at day 0 and day 42 would require an increase in “dose” of 158% (476 ± 21.6 mL of day 42 sRBCc was equipotent to 300 ± 0.0 mL of fresh RBCs). Employing a novel in vitro model of massive transfusion, L-TOF was used to quantify transfusion effectiveness by examining change in recipient blood L-TOF after receipt of either crystalloid (normal saline), sWB (citrate phosphate dextrose adenine-1), or WBAs. L-TOF enables direct comparison of O₂ delivery potency between different blood products, offering a predictive readout of transfusion quality.

Introduction

Blood transfusion is intended to correct anemia when it is believed to either impair adequate oxygen (O₂) delivery or when the resulting compensatory cardiovascular workload poses clinical risk. Current transfusion decisions are guided mainly by hemoglobin (Hb) concentration thresholds; therefore, quantitative assessment of anemia intolerance is primarily based on the assumption that Hb levels predict oxygenation. This assumption neglects differences in actual tissue O₂ delivery potential between a patient’s own red blood cells (RBCs) and transfused donor RBCs, or between different donor RBC preparations.^{1–4} As such, in addition to evaluating recipient tolerance to anemia severity, transfusion decision-making should include a blood product potency assay to inform O₂ delivery potential; however, we lack an accepted metric to guide this more nuanced assessment.

As transfusion medicine advances to incorporate alternatives to traditional blood products such as Hb-based O₂ carriers (HBOCs), especially given the significant variation in their O₂ binding characteristics relative to each other⁵ and to RBCs, it is imperative to evaluate the ability of these materials to not only successfully load but also deliver O₂. To achieve this aim, we developed an in vitro metric to predict the ability of RBCs, whole blood (WB), and artificial O₂ carriers to bind and release O₂ (ie, a metric for O₂ delivery potency). Potency

is defined by a drug's quantified biological activity and is expressed as a function of drug dose, whereas efficacy refers to the maximal achievable effect that a given drug formulation affords. If we consider Hb as a drug formulated in RBCs and define O₂ delivery (transport of O₂ from the lung-to-tissue) as the primary pharmacodynamic effect of Hb contained within a delivery system (be that in fresh WB [fWB], stored RBC concentrate [sRBCc], stored WB [sWB], or HBOC), we can begin to define the O₂ delivery potency of various "Hb formulations."

To predict O₂ delivery potency, we integrated data from matched O₂ association curves (OAC; high pH, the lungs) and O₂ disassociation curves (ODCs; low pH, tissue), simulating O₂ uptake in the lung and delivery to tissue across the range of physiologic O₂ gradients encountered during circulatory transit. Data from O₂ content curves derived from matched OAC and ODC were used to calculate lung-to-tissue O₂ flux (L-TOF), which we propose as a quantifiable predictive measure of O₂ delivery potency. Herein, we validate L-TOF as a metric for O₂ delivery potency by comparing fWB (day 0 storage) with sRBCc in additive solution 1 (AS-1) at 42 days of refrigerated storage. To test the model, we applied L-TOF as a metric for O₂ delivery effectiveness by simulating acute severe blood loss and massive transfusion. This study describes a novel method to define the O₂ delivery potency of RBCs, WB, and Hb-based O₂ delivery systems, filling a knowledge gap by offering a predictive readout of transfusion quality.

Materials and methods

Patients

This study was approved by the University of Maryland Baltimore human research ethics committee (HP-00104380) and written informed consent was obtained from all participants in accordance with the Declaration of Helsinki.

Transfusion units

In-date sRBCc units (AS-1) were obtained from the American Red Cross (The Holland Laboratory, Rockville, MD). sWB units (citrate phosphate dextrose adenine [CPDA-1]; University of Pittsburgh; institutional review board no. 22080148) were studied at day 26 ± 2 days, reflecting the median storage duration for in-theater WB transfusion during US military operations, 2015 to 2019, according to Jason Corley (US Army, personal written communication, 1 June 2022). Expired sRBCc (Vitalant Blood Center, Scottsdale, AZ, or ImpactLife, Davenport, IA) were used to purify Hb^{6,7} for O₂ delivery testing and to formulate ErythroMer (EM) or polymerized Hb (PolyHb). All units were stored at 4° C under standard blood-banking conditions.

Blood sampling

Fresh venous WB was drawn into heparin vacutainers and immediately placed on ice.

EM formulation

EM was fabricated using a lyophilized lipid cake containing phospholipids, cholesterol, and KC1003, a proprietary lipid of KaloCyte, as previously described.⁸

PolyHb formulation

Tense quaternary state PolyHb was manufactured as previously described.⁷

WBA formulations

WB analogs (WBAs) comprised (1) freeze-dried plasma (EZPlaz, Teleflex, Inc),⁹ (2) a synthetic platelet mimetic (SynthoPlate, Haima Therapeutics, Inc),^{10,11} and (3) 1 of 2 HBOCs, either polymerized T-state Hb (PolyHb)^{12–14} or a biosynthetic artificial RBC formulation, EM (KaloCyte Inc).^{15,16} The WBAs were formulated (volume per volume) as 55% freeze-dried plasma, 45% HBOC, and <1% SynthoPlate. The WBA was formulated at 2 different (final) HBOC Hb concentrations, either 2.9 g/dL (450 μ M) or 4.4 g/dL (675 μ M).

In vitro HMT model

To predict effectiveness of posttransfusion O₂ delivery for sWB and HBOCs, we developed a simple in vitro human massive transfusion (HMT) model to simulate the effect of a 50% or 75% transfusion on O₂ transport. Volumes of fWB (recipient) and sWB or HBOCs (as a component of WBA compositions) were mixed in 1:1 and 1:3 ratios to simulate blood volume replacements of 50% or 75%, respectively. fWB was also mixed at the same ratios with normal saline (NS) to study the dilutional effect resulting from crystalloid infusion, alone.

Statistical analyses

Results are presented as mean $\pm$ standard deviation or standard error of the mean, as indicated. Column statistics were performed, and a normality/lognormality test (Shapiro-Wilk test) was under-taken to confirm normal data distribution. Data were plotted as box and whisker plots, with each data point shown (with the box extending from the 25th to 75th percentile and whiskers representing minimum and maximum points). For parametric data, group means (2 groups) were analyzed using the Student *t* test; for >2 groups, means were compared by 1-way analysis of variance. Sample sizes and statistical test details can be found in the figure legends (Prism, GraphPad Inc; La Jolla, CA). A Pearson product moment correlation was computed to assess the relationship between different measurements (ie, partial pressure at which hemoglobin is 50% saturated with oxygen [P₅₀], L-TOF, etc). A *P* value <.05 was considered significant.

Results

Development of O₂ delivery potency metric

RBCs support adaptive homeostatic control of O₂ transport¹⁷ by modulating O₂ affinity in a fashion that optimizes both O₂ capture and release during physiologic stress.^{18–20} This ability is based on 3 key features of RBC physiology (ie, Hb cooperativity, HbO₂ affinity, and Hb concentration) and varies as a function of Hb formulation (eg, donor variability, processing, storage solution, storage duration, and blood product type). For additional information, refer to supplemental Results and supplemental Figure 1.

Calculating O₂ delivery potency: L-TOF

ODCs and OACs were determined²¹ (HEMOX analyzer, TSC Scientific Corporation, New Hope, PA); the O₂ loading curve (OAC) measured at high pH (~7.6) was used to simulate RBC pulmonary transit and the O₂ unloading curve (ODC), pH of ~6.8, to simulate RBC systemic transit under acidic (pH ~7.2) or ischemic (pH ~6.8) conditions (Figure 1A). To obtain higher resolution curves, the OAC/ODC curves were modeled using the Hill equation (supplemental Results; supplemental Equation 1). The OACs/ODCs were converted to O₂ content curves indexed to either Hb (O₂ flux per gram Hb, which quantifies theoretical Hb O₂ delivery potency; Figure 1B) or blood volume (O₂ flux per liter of blood, which accounts for Hb concentration in the blood as well as the theoretical Hb O₂ delivery potency; Figure 1C). From these curves we calculated L-TOF, our measure of O₂ delivery potency (L-TOF was determined both per gram Hb and per liter of blood). This was achieved by first determining the value of O₂ content per gram Hb (Figure 1B,D) or per liter of blood (Figure 1C,E) at a partial pressure of O₂ (pO₂) of 100 mm Hg from the OAC, which represents Hb or blood O₂ content after pulmonary transit. From this value, we subtracted all Hb or blood O₂ content values from the ODC at pH 7.2 (for study of RBCs alone in a simple transfusion) or at pH 6.8 (for study of recipient blood after simulated transfusion with either NS, sWB, or WBAs in our model of human massive transfusion; HMT), as follows:

$$L - TOF = O_{2contOAC} - O_{2contODC}$$

$$pO_2 \text{ delta} = 100 \text{ mmHg} - pO_{2ODC}$$

where, L-TOF is the flux of O₂ from the lungs to tissue, O_{2contOAC} is the O₂ content (per gram Hb or liter of blood) after pulmonary transit (at a pO₂ of 100 mm Hg) from the OAC, O_{2contODC} is the O₂ content (per gram Hb or liter of blood) in the tissue from the ODC, pO₂ delta is the change in pO₂ from the lungs to tissue, 100 mm Hg is the lung pO₂, and pO_{2ODC} is the tissue pO₂. This difference quantifies the full range of O₂ release across all potential lung-to-tissue O₂ gradients (Figure 1B–C). We then plotted pO₂ delta (x-axis) against the corresponding L-TOF delta (y-axis; Figure 1D–E). Finally, we integrated the total L-TOF for Hb, L-TOF(Hb), or blood, L-TOF(B), over the entire physiologic O₂ gradient, by quantifying the area under the curve (Figure 1D–E). Integration was performed using a trapezoidal method through the “pracma” package in R.

O₂ delivery potency and effectiveness of fWB (day 0) vs sRBCc (AS-1, day 42)

In comparison with fWB, sRBCc demonstrated significantly lower P₅₀ (higher O₂ affinity; ie, left-shifted ODC) at all 3 pH values ($P < .0001$ for all). Bohr plots that describe O₂ binding as a function of pH showed identical slopes for fWB and sRBCc (-0.438 ± 0.01 standard error of the mean (SEM) vs -0.3793 ± 0.03 SEM; fWB vs sRBCc; $P = .0952$; Figure 2A–B), indicating no difference in pH responsiveness between these Hb formulations (ie, fWB vs sRBCc).

ODCs for fWB and sRBCc (O₂ loading at pH 7.6 and O₂ unloading at pH 7.2; Figure 2C–D, respectively) demonstrate the impact of storage on O₂ capture and release by sRBCc (day 42 ± 1 day; AS-1). L-TOF plots for these Hb formulations (Figure 2E–F) quantify the resulting effect on blood O₂ delivery across the full physiologic O₂ gradient. L-TOF(B) plotted against blood Hb concentration highlights the significantly reduced potency and efficacy of sRBCc compared with fWB (Figure 2K). We calculated that, as a function of storage duration from day 0 to 42, L-TOF fell from 43.1 ± 1.4 to 27.2 ± 1.2 mL O₂ • mm Hg/g Hb, which is a theoretical potency loss of 37%, resulting in an effective dose reduction from 75.0 ± 0.0 to 47.3 ± 2.1 grams of Hb. Consequently, to achieve an O₂ delivery hypothetically equipotent to fWB, sRBCc at the end of its shelf life (ie, day 42) would require an increase in RBC dose of ~160%, from 1.0 ± 0.0 to 1.59 ± 0.07 units, or an increase in transfusion volume from 300 ± 0.0 to 476 ± 21.6 mL (Table 1).

O₂ delivery potency and effectiveness of fWB (day 0) vs sWB (CPDA-1, day 26 ± 2 days)

Table 2 shows the P₅₀ comparison at pH 6.8, 7.4, and 7.6 between fWB, sWB, and cell-free Hb. fWB demonstrated the highest P₅₀ at each pH, cell-free Hb the lowest, and that of sWB was between the aforementioned values (at each pH, all samples differed significantly from one another; $P < .0001$, analysis of variance). Bohr plots that describe O₂ binding as a function of pH (Figure 2B) showed significantly different slopes between fWB vs sWB (-0.438 ± 0.01 SEM vs -0.374 ± 0.01 SEM; fWB vs sRBCc; $P = .0044$) and fWB vs cell-free Hb (-0.438 ± 0.01 SEM vs -0.359 ± 0.02 SEM; fWB vs sRBCc; $P = .0244$), indicating different pH responsiveness between these Hb formulations.

We used L-TOF to demonstrate the impact of increased HbO₂ affinity of sWB compared with fWB upon transfusion efficacy (Figure 2G–H, respectively). L-TOF plots for these Hb formulations (Figure 2I–J) quantify the resulting effect on blood O₂ delivery across the full physiologic O₂ gradient. L-TOF(B) plotted against blood Hb concentration highlights the reduced O₂ delivery potency and effectiveness of sWB compared with fWB (Figure 2L). We determined that between sWB storage day 0 and day 26 ± 2 days, L-TOF fell from 62.9 ± 1.2 to 38.9 ± 4.0 mL O₂•mm Hg/g Hb, a relative theoretical O₂ delivery potency reduction of 38% (supplemental Table 1). Interestingly, the relative impact of storage upon sRBCc (pH 7.2 and 7.6) and sWB (pH 6.8 and 7.6) L-TOF overlay one another (Figure 2K–M).

Simulation of a massive crystalloid transfusion

We developed a simple benchtop human massive transfusion (HMT) model to evaluate the effect of a 50% or 75% volume transfusion on Hb and blood O₂ delivery potency. Dilution of fWB in NS at 1:1 and 1:3 ratios had no impact on HbO₂ affinity; as expected, the Hb O₂ content curves (milliliters of O₂ per gram Hb; Figure 3A) and the L-TOF Hb curves (Figure 3B) demonstrate that O₂ delivery potency of the dilutions was identical to that of fWB (Figure 3C). However, dilution of fWB by NS alters Hb concentration and thus blood O₂ transport capacity. This is demonstrated by the proportional reduction in blood O₂ content curves (milliliters of O₂ per liter of blood; Figure 3D) and the L-TOF blood curves (Figure 3E–F).

Simulation of a massive sWB transfusion

Blood storage causes depletion of Hb allosteric effectors (2,3-diphosphoglycerate [2,3-DPG] and ATP), increasing HbO₂ affinity and shifting the ODC left,²² as reflected in the sWB (day 26 ± 2 days) Bohr plot (Figure 2A). Mixing fWB with sWB at 1:1 and 1:3 ratios increased HbO₂ affinity of the mixtures compared with fWB alone, resulting in a leftward shift of the Hb O₂ content curves (Figure 4A). Consequently, mixtures of sWB with fWB demonstrated depressed L-TOF Hb curves (Figure 4B), illustrating the impact of storage upon transfusion effectiveness regarding O₂ delivery (Figure 4C). In fact, O₂ delivery potency for neat sWB (day 26 ± 2 days) was more like that of cell-free Hb than fWB (38.9 ± 4.0 vs 27.6 ± 0.12 vs 62.9 ± 1.2 mL O₂•mm Hg/g Hb; sWB vs cell-free Hb vs fWB, respectively). Because we normalized the Hb concentration of fWB and sWB at the outset of the experiment (to 11 g/dL) to exclude Hb concentration variance between blood donors, the blood O₂ content curves (milliliters of O₂ per liter of blood; Figure 4D–F) simply mirror the L-TOF Hb curves (milliliters of O₂ per gram Hb) by a factor of 110 (ie, the normalized Hb concentration per liter of blood).

Simulation of a massive WBA transfusion

WBAs comprising PolyHb and EM HBOCs were formulated at an Hb concentration of 450μM (2.9 g/dL), which was then mixed with fWB at 1:1 and 1:3 ratios. Relative to fWB, both WBA PolyHb (supplemental Figure 2A) and WBA EM (supplemental Figure 2G) had similar HbO₂ content curves, resulting in higher L-TOF curves for the WBAs than sWB (Figure 5A–B,E–F). Although PolyHb and EM are formulated with Hb from outdated transfusion units, polymerization of deoxyHb (for PolyHb) and incorporation of allosteric effector (for EM), result in Hb O₂ delivery potency that matches that of fWB. Consequently, mixtures of fWB with either WBA outperformed mixtures of fWB with sWB in terms of L-TOF(Hb) (milliliters of O₂ per gram Hb), as demonstrated in Figure 5B,F. However, because of the low Hb concentration of WBA PolyHb and WBA EM (450μM or 2.9 g/dL Hb before mixing with fWB), fWB:WBA mixtures contained a significantly lower Hb concentration than corresponding fWB:sWB mixtures. Consequently, both WBA mixtures demonstrated lower blood O₂ content curves than sWB mixtures (milliliters of O₂ per liter of blood; supplemental Figure 2D,J), which was reflected in lower L-TOF curves (milliliters of O₂ per liter of blood; Figure 5C,G), resulting in reduced effectiveness of blood O₂ delivery (milliliters of O₂ per liter of blood), compared with sWB at the same mixing ratios (Figure 5D,H). These data illustrate the importance of Hb dose as well as Hb formulation O₂ delivery potency, regarding transfusion effectiveness. Improvement in WBA formulation and performance in our HMT model and the resulting impact on transfusion parameters are shown in supplemental Table 2 (note EM version 3 has similar O₂ delivery potency to fWB).

Effect of increasing WBA HBOC Hb concentration on transfusion effectiveness

In our HMT model, WBA EM or WBA PolyHb formulated at higher Hb concentration (WBA Hb concentration of 450μM [2.9 g/dL] vs 675μM [4.4 g/dL]) significantly improved L-TOF(B) (milliliters of O₂ per liter of blood) at both the 1:1 (Figure 6A) and 1:3 (Figure 6B) mixing ratios; higher Hb concentration increases O₂ carrying capacity. Comparison is

made with fWB:NS, which indicates the effect if a crystalloid was transfused with no O₂ carrying capacity, and fWB:sWB, the current standard practice.

Modeled WBA HBOC Hb concentration to achieve O₂ delivery with equipotency to sWB

We used L-TOF(Hb) to model the WBA HBOC Hb concentration required to achieve theoretical O₂ delivery in L-TOF(B) that is equipotent to that of sWB in our 1:1 and 1:3 mixing experiments (supplemental Results).

Modulating EM O₂ delivery dynamics

We evaluated manipulation of EM P₅₀ by varying the ratio of Hb concentration to ATP concentration in the EM payload, upon formulation O₂ delivery potency. Hb:ATP at a 1:1 molar ratio resulted in an O₂ unloading P₅₀ (pH 6.8) of 39.2 mm Hg (supplemental Figure 3C) and an O₂ loading P₅₀ (pH 7.6) of 33.2 mm Hg (supplemental Figure 3C). Importantly, the HbO₂ saturation of the loading curve (pH 7.6) at a pO₂ of 100 mm Hg (ie, blood leaving the lungs) was 96.5%; however, at an Hb:ATP of 1:2, the O₂ unloading P₅₀ (pH 6.8, 50.1 mm Hg; supplemental Figure 3D) and O₂ loading P₅₀ (pH 7.6, 42.1 mm Hg; supplemental Figure 3D) increased substantially and reduced the pulmonary vein HbSO₂ to 90.9%, limiting arterial blood O₂ content. L-TOF(Hb) analysis of Hb O₂ delivery potency demonstrated that an Hb:ATP of 1:2 was superior to a 1:1 ratio, because the increase in O₂ release during systemic perfusion more than compensated for the reduction in O₂ capture during pulmonary perfusion (Figure 6C). Because Hb formulation (different P₅₀) influences the ability to capture O₂ during pulmonary transit, we estimated the impact of alveolar hypoxia upon L-TOF(Hb) for the Hb formulations alone (Figure 6D) and in our HMT model (Figure 6E). Notably, both HBOCs demonstrate superior or equivalent O₂ delivery potency to that of sWB across the full range of modeled alveolar hypoxia (100 → 50 mm Hg).

Comparison of P₅₀ and L-TOF

Finally, we performed correlation and Bland-Altman analysis (supplemental Figure 4), which indicates that P₅₀ and L-TOF(Hb) capture different information and that P₅₀ may be less informative than L-TOF(Hb) in the assessment of O₂ carriers when P₅₀ and O₂ delivery potency are estimated to be higher than that of fWB.

Discussion

Judicious blood product management and transfusion decision-making requires rigorous assessment of blood product quality.²³ Herein, we demonstrate use of data from a simple benchtop assay to quantify blood product O₂ delivery potency and predict posttransfusion effectiveness. This metric, termed L-TOF, captures and incorporates the 3 key aspects of Hb physiology that define blood O₂ delivery potential (ie, Hb cooperativity and allosteric, HbO₂ affinity, and Hb concentration). Moreover, L-TOF reflects the effect of Hb formulation upon ability to both capture O₂ in the lungs and release it to tissues, aligning with the principal therapeutic goal of transfusion (ie, to enhance tissue O₂ delivery). Furthermore, L-TOF quantifies potential O₂ transport across physiological and pathophysiologic O₂ and pH gradients, rather than at a single pO₂ and pH (such as P₅₀), incorporating information regarding Hb O₂ delivery potency of blood product formulations exposed to pragmatic

physiologic scenarios. Ultimately, blood product L-TOF quantification enables direct comparison of O₂ delivery potency between fWB, sRBCc (AS-1), sWB (CPDA-1), and HBOCs. Additionally, L-TOF can also be used to measure posttransfusion blood mixtures (as we show in our novel benchtop HMT model), enabling direct comparison of O₂ delivery efficacy between crystalloid/colloid, fWB, sRBCc, sWB, and HBOCs.

We acknowledge that L-TOF is a surrogate for actual O₂ delivery, for which there is no current direct metric, and is based on several assumptions. ODCs in vivo are tissue specific, each with unique pH and pO₂ gradients, whereas L-TOF considers only 2 curves at different pHs. Notably, the 2 pH values selected to acquire the OAC and ODC represent the extremes encountered during physiologic stress (however, we suggest that this is the condition for which blood product O₂ delivery potency and transfusion effectiveness should be evaluated). There are notable limitations to consider when applying this methodology. For example, ODCs reflecting maximal tissue exertion should be measured at temperatures higher (exercise) and lower (to consider high altitude, cardiopulmonary bypass, and organ perfusion before transplant) than 37°C.²⁴ We are also aware that in our assessment of O₂ flux, we integrate L-TOF across the complete gradient from 0 to 100 mm Hg pO₂ (ie, from no O₂ release to complete O₂ release). We acknowledge that the process of O₂ release from RBCs is regulated by surrounding tissue O₂ demand and even the most highly metabolic tissues never extract all O₂ from circulating RBCs.²⁵ Despite these assumptions, we believe that this novel approach adds a much needed means to quantify O₂ delivery potency, allowing direct comparison of blood product formulation and comparison of HBOC performance with that of crystalloid, colloid, and blood (fWB, sWB).

L-TOF introduces the concept of predictive O₂ delivery potency of blood products, which we suggest is long overdue. Herein, we present quantified theoretical loss of O₂ delivery potency over storage duration for RBCc (Table 1), demonstrating an end of storage (day 42) potency drop to 63% of that at day 0. This loss reflects well-characterized metabolic reprogramming that occurs during cold storage, leading to depletion of the critical allosteric effector 2,3-DPG. The consequent increase in Hb O₂ affinity impairs O₂ unloading and represents the principal mechanism underlying the hypothetically reduced O₂ delivery potency of stored RBCs. In contrast, storage-related shifts in O₂ binding are generally limited in HBOCs, because they lack cellular metabolism and are engineered with chemically fixed or structurally stabilized O₂ affinities. Additionally, future products are moving toward freeze drying, which should render O₂ delivery potency stable for months. Future studies will incorporate the impact of storage on HBOC O₂ delivery potency, because this is obviously an important safety consideration.

In vivo, the functional performance of HBOCs is primarily limited by oxidative processes that generate methemoglobin (metHb), ultimately resulting in their inability to transport O₂. Consequently, an important performance indicator for the various O₂ transporters currently under development is their functional in vivo half-life, the duration over which O₂ delivery capacity is maintained before oxidative inactivation. We are planning future studies to capture this pharmacodynamic parameter and to validate our L-TOF metric in vivo by correlating it with physiologic measures of oxygenation under controlled pharmacologic modulation of Hb O₂ affinity.

L-TOF, by quantifying the theoretical O₂ delivery potency of blood products, additionally allows us to consider blood product dosing analogous to potency-based dosing of conventional therapeutics. Currently, every therapeutic approved by the regulatory agencies has a dosing range that is identified to optimize therapeutic benefit and is directly derived from precisely quantified potency. As with all therapeutics, if an administered dose is outside the desired therapeutic window, either effectiveness will be suboptimal, and/or there will be an increase in undesired side effects. The same is true for blood products. Historically, transfusion decision-making has been based on specific threshold Hb concentrations, which focuses solely on circulating O₂ content, without consideration of the dramatic differences in O₂ delivery potency between recipient and donor RBCs or among various RBC formulations. As we strive to more clearly understand in vivo O₂ delivery dynamics of transfusion, perhaps dosing strategies and transfusion triggers can be adjusted to ensure that blood products are dosed to achieve a specific O₂ delivery potency (ie, dose optimization).^{26,27}

Our analysis of sRBCc (AS-1) and sWB (CPDA-1) highlights the impact of the storage lesion on the theoretical O₂ delivery potency.²⁸ sRBCc (AS-1) was tested at the end of shelf life (42 ± 1 days), whereas sWB was tested at 26 ± 2 days. It should be noted that sWB in CPDA-1 has a shelf life of 35 days and it is likely that O₂ delivery potency is even further affected at the end of sWB shelf life. Although it is thought that the RBC impairments observed with blood storage may reverse once the RBCs are transfused and circulating, the timeframe (which for 2,3-DPG studied ex vivo appears to be hours)²² and the true extent of RBC recovery in vivo remains to be determined. To address this, we are planning future experiments in which we will incubate sRBCc in fresh plasma (ex vivo) and quantify how quickly O₂ delivery potency and 2,3-DPG levels are restored. Nevertheless, at the point of transfusion and immediately after, when O₂ delivery is most crucial, L-TOF indicates that the theoretical O₂ delivery potency of sRBCc and sWB are significantly impaired compared with that of fWB. Consequently, to achieve O₂ delivery equipotent to that of fWB, sRBCc and sWB must be transfused at a higher dose, thus increasing the potential for adverse side effects (Table 1; supplemental Table 1).

Although L-TOF was designed for the assessment of blood storage products and HBOC development, to allow rapid identification of product features requiring manipulation to optimize O₂ delivery potency (ie, Hb allosteric, HbO₂ affinity, or Hb concentration), we propose that this metric could also yield essential information of native RBC physiologic function, perhaps in routine clinical practice, in which it could potentially be used as an important biomarker in terms of personalized medicine to assess blood health. This might be especially relevant in disease states, in which O₂ delivery homeostasis is likely impaired due to altered Hb cooperativity, HbO₂ affinity, and Hb concentration.²⁹

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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Key Points

- Hb-based transfusion dosing overlooks patient, donor, and preparation differences in actual O₂ delivery potency.
- L-TOF predicts and compares O₂ delivery potency across blood products, providing a transfusion quality metric.

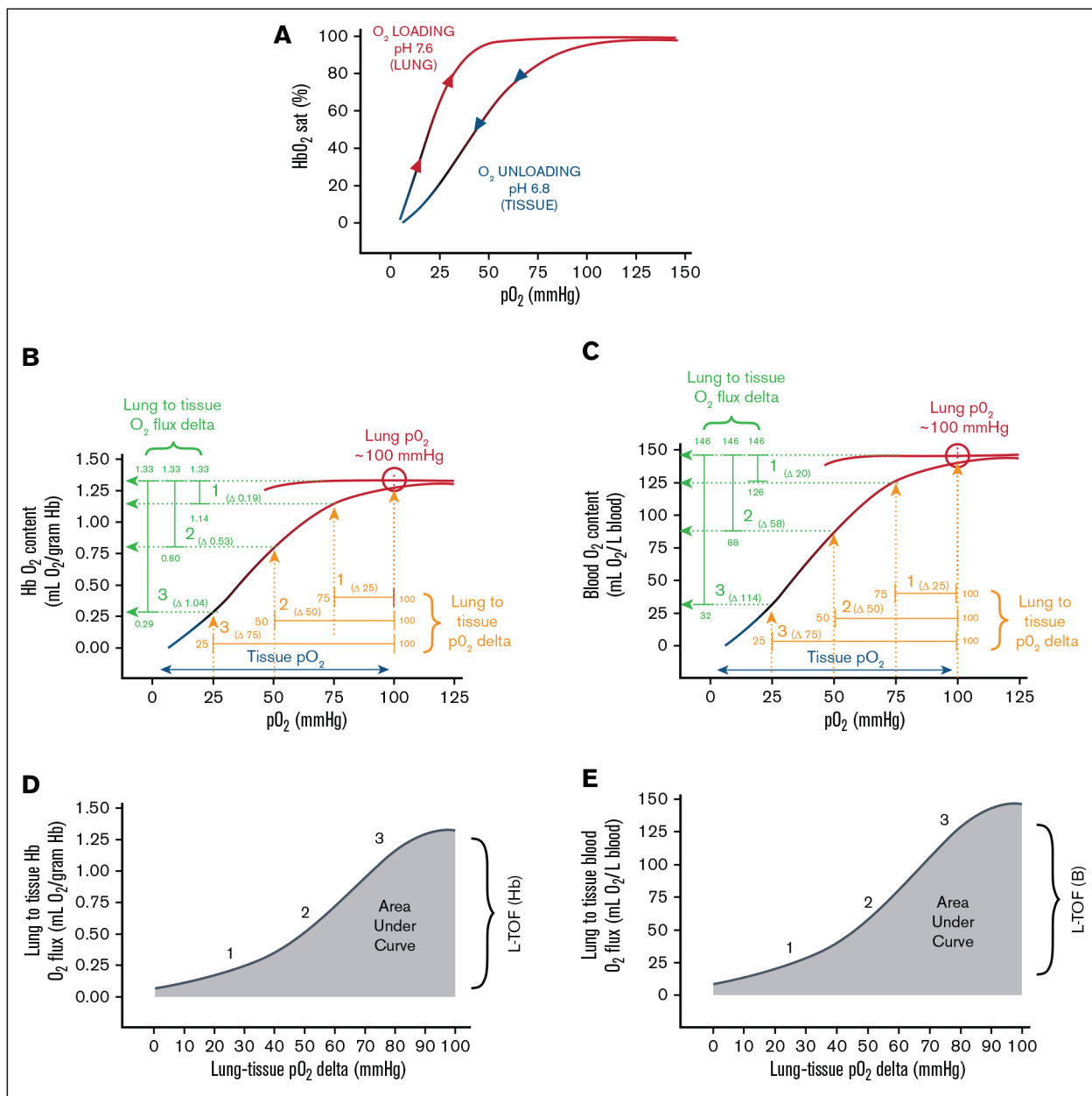


Figure 1. Estimating O₂ delivery potency of blood by quantifying L-TOF.

(A) OAC (ie, O₂ loading curve, pH ~7.6, lung perfusion; n = 9, without error bars) and ODC (ie, O₂ unloading curve, pH ~6.8, tissue perfusion; n = 14, without error bars) measured at 37°C. Mean curves converted to O₂ content, either as (B) Hb O₂ content (mL O₂/g Hb), or (C) blood O₂ content (mL O₂/liter blood; Hb concentration of 11 g/dL). Lung to tissue (D) Hb and (E) blood O₂ flux is calculated from the O₂ content curves by subtracting ODC values (0–99 mm Hg) from the OAC value at 100 mm Hg, reflecting lung O₂ loading minus maximal tissue unloading. The x-axis lung-to-tissue pO₂ delta represents potential lung-to-tissue pO₂ gradients; y-axis values represent the absolute amount of O₂ delivered for any given pO₂ gradient. L-TOF values plotted as (D) mL of O₂ delivered per gram Hb for L-TOF(Hb), or (E) mL of O₂ delivered per liter of blood for L-TOF(B). Points 1 through 3

in panels B-C are labeled in panels D-E for orientation. Area under the curve (AUC) defines total (D) L-TOF(Hb), or (E) L-TOF(B). Note in this figure, the mean curves generated in panel A have been used to generate panels B-E. At all other instances, L-TOF is generated from single O₂ loading and O₂ unloading curves from the same sample. L-TOF(Hb), L-TOF per gram of Hb; L-TOF(B), L-TOF per liter of blood.

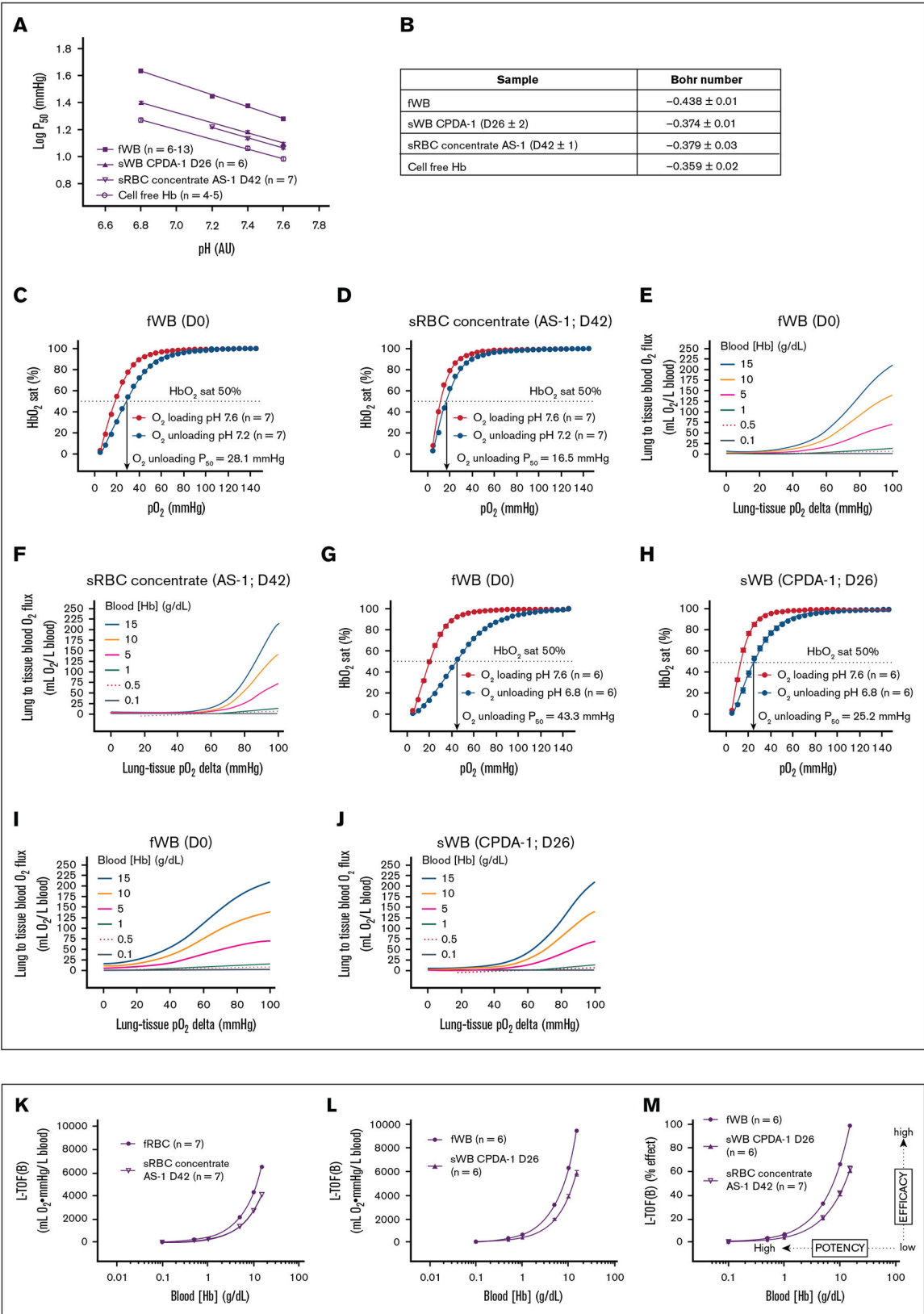
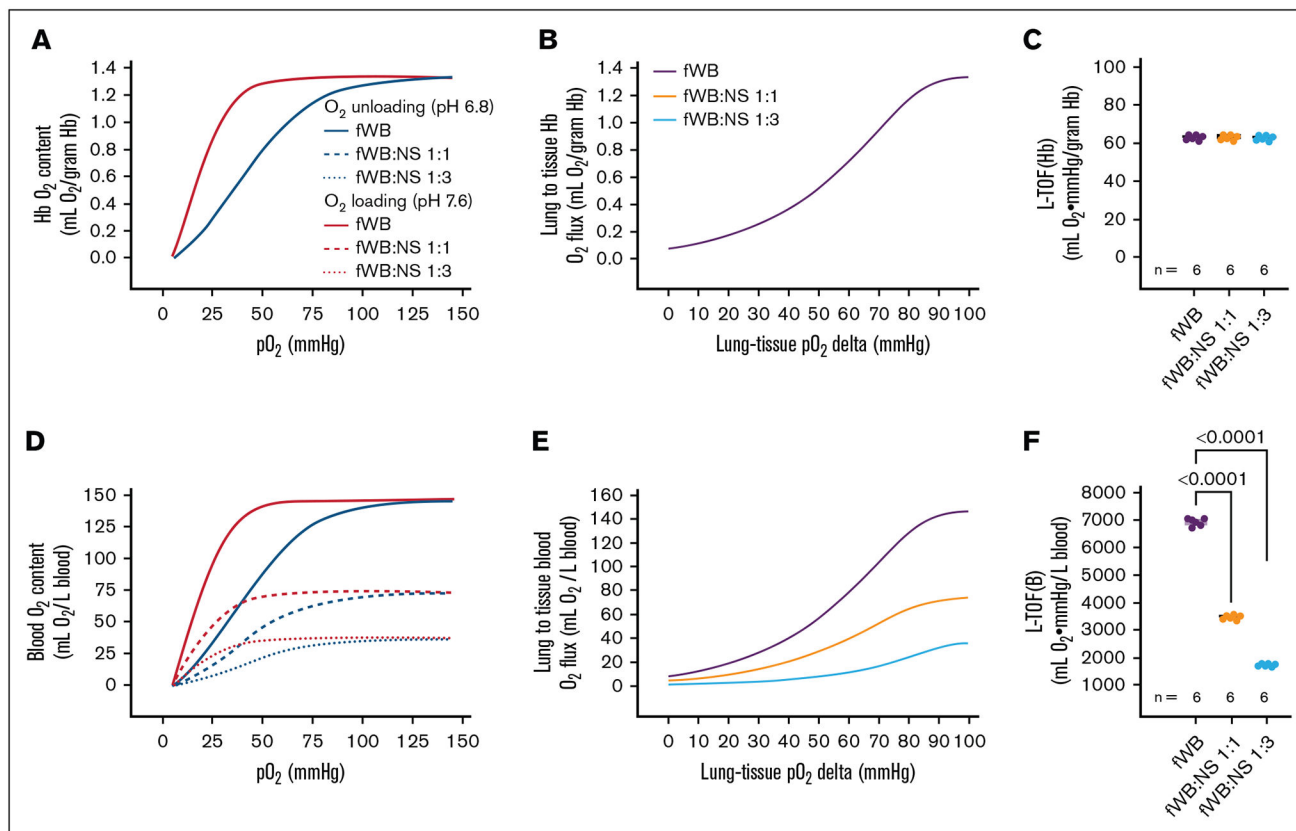


Figure 2. Blood O₂ delivery potency and efficacy; comparison of fWB, stored blood products, and cell-free Hb.

(A) Bohr plots from ODCs measured in fWB (day 0; n = 6–13), sRBCc (AS-1, day 42 ± 1 day; n = 7), sWB (CPDA-1, day 26 ± 2 days; n = 6), and cell-free Hb at 37°C pH 6.8–7.6 (n = 4–5). (B) Mean Bohr values (±SEM). (C) OAC (pH 7.6, the lung; n = 7; panel G, n = 6) and ODCs (pH 7.2 or 6.8, routine or ischemic tissue hypoxia) shown for fWB (panel C; n = 7; panel G, n = 6), sRBCc (panel D; n = 7), or sWB (panel H; n = 6). OAC blood O₂ content at pO₂ of 100 mm Hg minus ODC values (at either pH 7.2 or 6.8) gives lung-to-tissue O₂ content delta. This delta is plotted against lung-to-tissue pO₂ delta and multiplied by different Hb concentrations (0.1–15 g/dL), to obtain blood Hb concentration–dependent O₂ flux curves for (panel E, n = 7; panel I, n = 6, both plotted without error bars) fWB, (F) sRBCc (n = 7, plotted without error bars), or (J) sWB (n = 6, plotted without error bars). AUC of each dose gives (panel K, n = 7; panel L, n = 6) L-TOF(B) (mL O₂•mm Hg/L blood), from which (M) percentage (%) effect of L-TOF(B) calculated, showing greater potency and efficacy of fWB in comparison with sRBCc or sWB. L-TOF(B), lung to tissue O₂ flux per liter of blood.



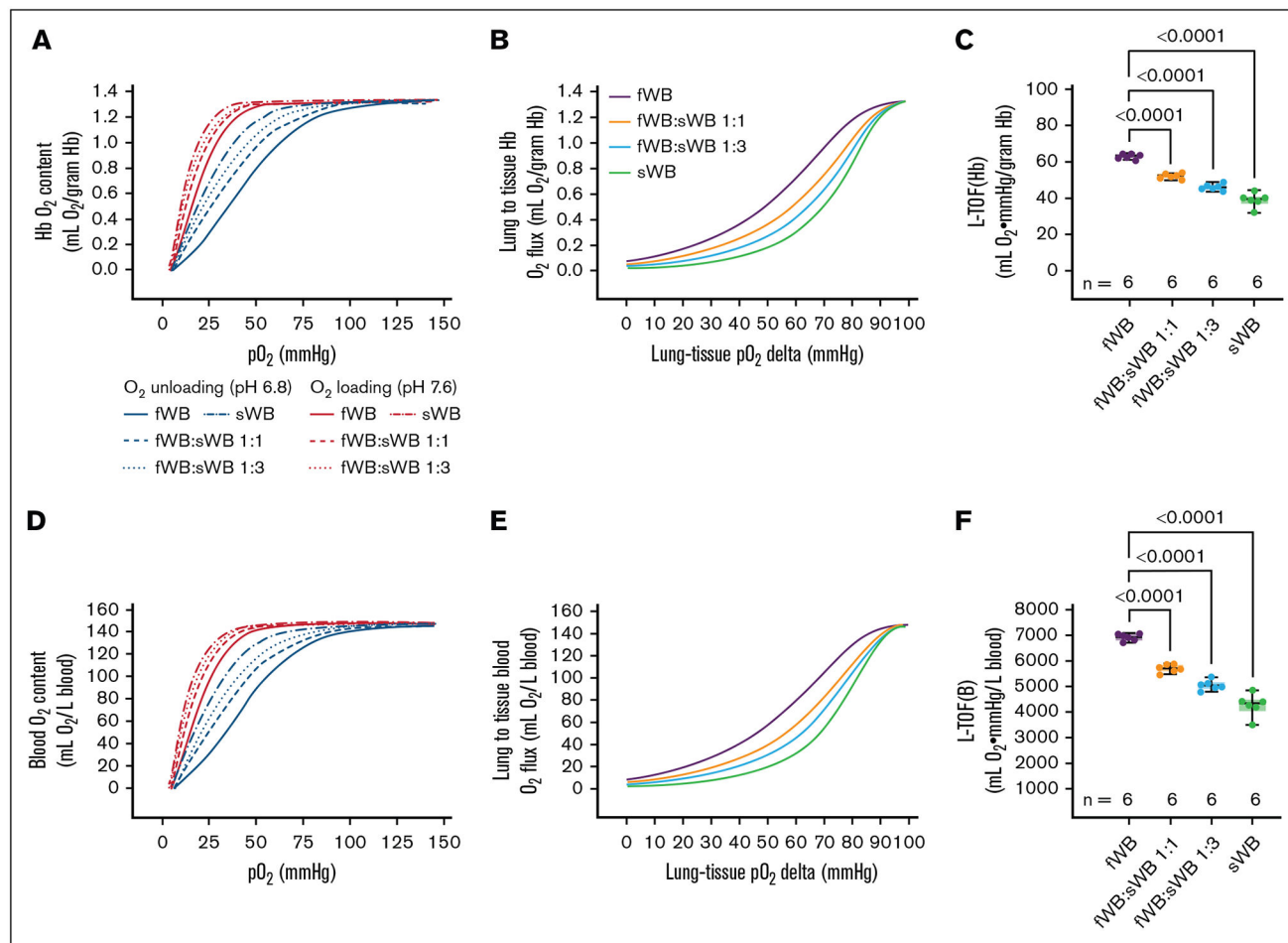


Figure 4. Simulation of a massive sWB transfusion on Hb and blood O₂ delivery dynamics.

(A) Hb O₂ content curves (mL O₂/g Hb) for fWB, fWB:sWB 1:1, fWB:sWB 1:3, and sWB (n = 6 for all). (B) Corresponding L-TOF Hb O₂ flux curves (mL O₂/g Hb, n = 6 for all). (C) L-TOF(Hb) (mL O₂•mm Hg/g Hb, n = 6 for all). (D) Blood O₂ content curves (mL O₂/L blood) for fWB, fWB:sWB 1:1, fWB:sWB 1:3, and sWB (n = 6 for all). (E) Corresponding L-TOF blood O₂ flux curves (mL O₂/L blood, n = 6 for all) and (F) L-TOF(B) (mL O₂•mmHg/L blood, n = 6 for all). O₂ content curves for panels A,D and O₂ flux curves for panels B,E plotted as mean without error bars. Box-and-whiskers plot in panels C,F plotted with each data point shown, the box extends from the 25th to 75th percentile, and whiskers represent minimum and maximum points. Group differences were analyzed using 1-way ANOVA followed by pairwise comparisons (fWB vs each group) with the Dunnnett's post hoc test. L-TOF(Hb), L-TOF per gram of Hb; L-TOF(B), L-TOF per liter of blood.

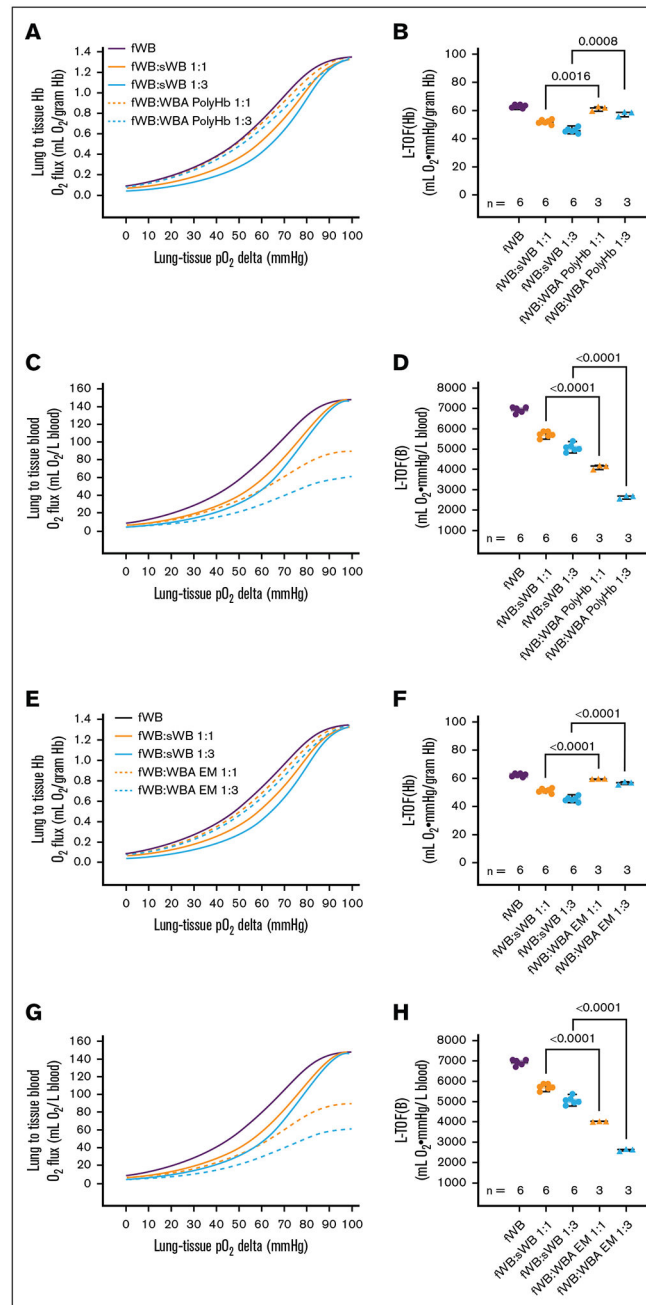


Figure 5. Simulation of a massive transfusion using WBAs with PolyHb or EM.

(A) L-TOF Hb O₂ flux curves (mL O₂/g Hb) for fWB (n = 6), fWB:sWB (1:1 and 1:3, n = 6 for both), and fWB:WBA PolyHb (1:1 and 1:3; PolyHb at 1mM Hb concentration before mixing to formulate the WBA, n = 3 for both). (B) L-TOF(Hb) (mL O₂•mm Hg/g Hb). (C) L-TOF blood O₂ flux curves (mL O₂/L blood) for fWB (n = 6), fWB:sWB (1:1 and 1:3, n = 6 for both), and fWB:WBA PolyHb (1:1 and 1:3; PolyHb at 1mM Hb concentration before mixing to formulate the WBA, n = 3 for both). (D) L-TOF(B) (mL O₂•mm Hg/L blood). (E) L-TOF Hb O₂ flux curves (mL O₂/g Hb) for fWB (n = 6), fWB:sWB (1:1 and 1:3, n = 6 for both), and fWB:WBA EM (1:1 and 1:3; EM at 1mM Hb concentration before mixing

to formulate the WBA, $n = 3$ for both). (F) L-TOF(Hb) ($\text{mL O}_2 \cdot \text{mm Hg/g Hb}$). (G) L-TOF blood O_2 flux curves ($\text{mL O}_2/\text{L blood}$) for fWB ($n = 6$), fWB:sWB (1:1 and 1:3, $n = 6$ for both), and fWB:WBA EM (1:1 and 1:3; EM at 1mM Hb concentration before mixing to formulate the WBA, $n = 3$). (H) L-TOF(B) ($\text{mL O}_2 \cdot \text{mm Hg/L blood}$). O_2 flux curves for panels A, C, E, G plotted as mean without error bars. Box-and-whiskers in panels B, D, F, H plotted with each data point shown, the box extends from the 25th to 75th percentile, and whiskers indicate minimum and maximum points. Differences between dilution ratios (1:1 vs 1:3) were analyzed using unpaired 2-tailed t tests for each product type (fWB:sWB vs fWB:WBA PolyHb or fWB:WBA EM). L-TOF(Hb), L-TOF per gram of Hb; L-TOF(B), L-TOF per liter of blood.

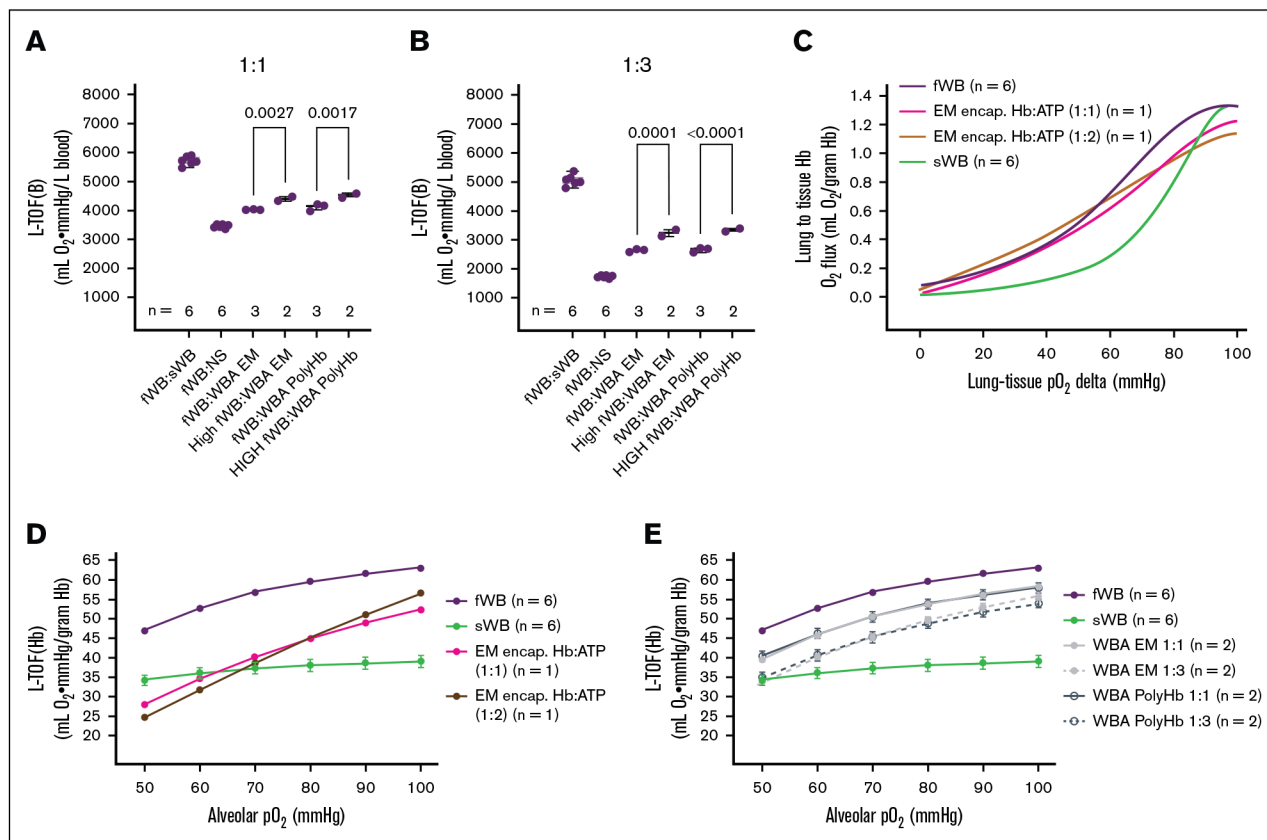


Figure 6. Modulating O₂ delivery dynamics of artificial blood substitutes.

Increasing HBOC Hb concentration before WBA formulation and fWB mixing improved blood O₂ delivery potency by raising O₂ carrying capacity. (A) L-TOF(B) (mL O₂•mmHg/L blood) at the 1:1 ratio for fWB mixed with sWB (n = 6), NS (n = 6), WBA EM (n = 3), HIGH WBA EM (n = 2), WBA PolyHb (n = 3), HIGH WBA PolyHb (n = 2; EM and PolyHb formulated at 1mM or ~1.5mM Hb concentration; HIGH = 1.5mM before mixing to formulate the WBA). (B) L-TOF(B) (mL O₂•mmHg/L blood) at 1:3 ratio (same groups as in panel A). (C) EM performance tuned by altering ratio of internal Hb concentration to allosteric effector concentration, with single component analysis (fWB, sWB, or EM alone) showing effect of the ratio of internal Hb concentration to ATP concentration. L-TOF Hb O₂ flux curves (mL O₂•mmHg/g Hb) for fWB (n = 6), sWB (n = 6), or EM at varying ratios of Hb concentration to ATP concentration (n = 1). (D) L-TOF(Hb) (mL O₂•mmHg/g Hb) vs alveolar pO₂ (mmHg), impact of alveolar hypoxia. (E) Same comparison including WBA EM (n = 2) and WBA PolyHb (n = 2). Box and whisker plots in panels A-B show individual points, the box extends from the 25th to 75th percentile, whiskers show the minimum and maximum points. Curves in panel C show means without error bars; panel D-E show mean $\pm$ SEM. Group differences were evaluated by 1-way ANOVA across all 6 groups, followed by 2 prespecified pairwise contrasts (WBA EM vs WBA EM HIGH; WBA PolyHb vs WBA PolyHb HIGH). Multiple comparison *P* values for these contrasts were adjusted using the Šidák method. L-TOF(B) = lung to tissue O₂ flux per liter of blood.

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Table 1. Comparison of O₂ delivery potency as a function of RBCc storage duration (AS-1), with theoretical impact on transfusion parameters

Storage duration (d)	L-TOF (potency; mL O ₂ /mm Hg/g Hb)	Relative theoretical potency	Effective theoretical dose (g Hb)	Equipotent RBC units	Equipotent Tx volume (mL)
0	43.1 ± 1.4	1.00 ± 0.00	75.0 ± 0.0	1.00 ± 0.00	300 ± 00.0
4 ± 0	45.3 ± 1.6	1.05 ± 0.04	78.7 ± 2.7	0.95 ± 0.03	286 ± 10.0
7 ± 1	37.7 ± 4.6	0.87 ± 0.11	65.6 ± 8.0	1.16 ± 0.14	348 ± 41.6
14 ± 1	31.6 ± 2.4	0.73 ± 0.06	54.9 ± 4.2	1.37 ± 0.10	412 ± 30.4
21 ± 1	29.0 ± 1.6	0.67 ± 0.04	50.4 ± 2.8	1.49 ± 0.08	448 ± 24.6
28 ± 1	29.8 ± 2.1	0.69 ± 0.05	51.8 ± 3.6	1.45 ± 0.10	436 ± 29.0
35 ± 1	28.0 ± 0.5	0.65 ± 0.01	48.6 ± 0.9	1.54 ± 0.03	463 ± 07.9
42 ± 1	27.2 ± 1.2	0.63 ± 0.03	47.3 ± 2.1	1.59 ± 0.07	476 ± 21.6

Tx, transfusion.

Table 2.P₅₀ for fWB, sRBCc, sWB, and cell-free Hb

Sample	P ₅₀ (mm Hg)			
	pH 6.8	pH 7.2	pH 7.4	pH 7.6
fWB (n = 6–13)	43.3 ± 1.0	28.1 ± 1.2	24.0 ± 1.8	19.1 ± 0.9
sWB (n = 6)	25.2 ± 1.4		15.3 ± 0.8	12.5 ± 0.5
sRBCc (n = 7)		16.5 ± 0.6	13.8 ± 0.8	11.7 ± 0.7
Cell-free Hb (n = 4–5)	17.6 ± 0.4		10.5 ± 0.4	8.8 ± 0.6

Data are shown as mean ± standard deviation. Samples measured at 37° C in Bis-Tris 50mM, NaCl 100mM. At each pH, all samples significantly different from one another ($P < .0001$; Student *t* test).